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CASE H_400_baseline: Point H - 400 + 800 MHz (baseline)
layout (QUI -> RM): 44x 800 MHz + 33x 400 MHz

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--- Lines C/D: 4 K supply / return ---

C: D = 134.5 mm (DN125, EN OD 139.7 x 2.6) m QUI = 1955.9 g/s dP = 64.626 mbar v_max = 1.07 m/s
D: D = 213.9 mm (DN200, EN OD 219.1 x 2.6) m QUI = 1918.9 g/s dP = 33.212 mbar v_max = 2.50 m/s

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Line C key nodes

Node	idx	x [m]	P [bar]	T [K]	h [J/g]	m [g/s]	dm_node [g/s]	v [m/s]	rho [kg/m^3]	flash [%]
RM	78	1039.37	3.000000	4.6000	12.2590	0.0000	0.0000	0.0000	127.5863	-
RM-1	77	1027.04	3.000000	4.6000	12.2590	58.9113	58.9113	0.0325	127.5863	3.34
RM-2	76	1014.71	3.000001	4.5942	12.2303	117.7309	58.8196	0.0649	127.7052	3.19
400 first	45	632.40	3.010915	4.5757	12.1425	1934.7415	58.5400	1.0628	128.1311	2.72
800 last	44	619.16	3.012029	4.5754	12.1416	1935.2215	0.4800	1.0629	128.1414	-
MID	39	552.98	3.017602	4.5742	12.1369	1937.6214	0.4800	1.0638	128.1925	-
QUI+2	2	63.24	3.059209	4.5651	12.1025	1955.3812	0.4800	1.0704	128.5705	-
800 first	1	50.00	3.060343	4.5648	12.1015	1955.8611	0.4800	1.0706	128.5807	-
QUI	0	0.00	3.064626	4.5639	12.0980	1955.8611	0.0000	1.0703	128.6192	-

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Line D key nodes

Node	idx	x [m]	P [bar]	T [K]	h [J/g]	m [g/s]	dm_node [g/s]	v [m/s]	rho [kg/m^3]	flash [%]
RM	78	1039.37	1.300000	4.6000	31.5408	0.0000	0.0000	0.0000	20.5229	-
RM-1	77	1027.04	1.300000	4.4977	30.4696	58.4313	58.4313	0.0737	22.0723	-
RM-2	76	1014.71	1.300000	4.4999	30.4943	116.7709	58.3396	0.1475	22.0339	-
400 first	45	632.40	1.294420	4.4963	30.5177	1918.9017	58.0600	2.4380	21.9033	-
800 last	44	619.16	1.293849	4.4958	30.5193	1918.9017	0.0000	2.4393	21.8911	-
MID	39	552.98	1.290990	4.4936	30.5274	1918.9017	0.0000	2.4461	21.8304	-
QUI+2	2	63.24	1.269583	4.4770	30.5871	1918.9017	0.0000	2.4979	21.3784	-
800 first	1	50.00	1.268999	4.4766	30.5887	1918.9017	0.0000	2.4993	21.3661	-
QUI	0	0.00	1.266788	4.4748	30.5948	1918.9017	0.0000	2.5047	21.3197	-

--- Lines A/B: 2 K supply / return ---

A: D = 84.9 mm (DN80, EN OD 88.9 x 2.0) m QUI = 513.4 g/s dP = 13.463 mbar v_max = 0.61 m/s
B: D = 347.6 mm (DN350, EN OD 355.6 x 4.0) m QUI = 513.4 g/s dP = 1.790 mbar v_max = 7.80 m/s

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Line A key nodes

Node	idx	x [m]	P [bar]	T [K]	h [J/g]	m [g/s]	dm_node [g/s]	v [m/s]	rho [kg/m^3]	flash [%]
RM	78	1039.37	-	-	-	-	0.0000	-	-	-
RM-1	77	1027.04	-	-	-	-	0.0000	-	-	-
RM-2	76	1014.71	-	-	-	-	0.0000	-	-	-
400 first	45	632.40	1.300000	2.4500	4.5568	0.0000	0.0000	0.0000	147.7382	-
800 last	44	619.16	1.300000	2.4500	4.5568	11.9437	11.9437	0.0143	147.7382	12.53
MID	39	552.98	1.300022	2.3081	4.2295	70.9445	11.7557	0.0845	148.3219	11.13
QUI+2	2	63.24	1.309927	2.2026	3.9327	501.7812	11.5903	0.5964	148.6273	9.86
800 first	1	50.00	1.310640	2.2015	3.9293	513.3696	11.5884	0.6101	148.6303	9.84
QUI	0	0.00	1.313463	2.1976	3.9167	513.3696	0.0000	0.6101	148.6415	-

Line B key nodes										
Node	idx	x [m]	P [bar]	T [K]	h [J/g]	m [g/s]	dm_node [g/s]	v [m/s]	rho [kg/m^3]	flash [%]
RM	78	1039.37	-	-	-	-	0.0000	-	-	-
RM-1	77	1027.04	-	-	-	-	0.0000	-	-	-
RM-2	76	1014.71	-	-	-	-	0.0000	-	-	-
400 first	45	632.40	0.031000	2.0000	25.0279	0.0000	0.0000	0.0000	0.7872	-
800 last	44	619.16	0.031000	1.9968	25.0279	11.9437	11.9437	0.1599	0.7872	-
MID	39	552.98	0.030997	2.0900	25.5781	70.9445	11.7557	1.0008	0.7470	-
QUI+2	2	63.24	0.029689	2.1038	25.6813	501.7812	11.5903	7.4600	0.7088	-
800 first	1	50.00	0.029593	2.1035	25.6817	513.3696	11.5884	7.6573	0.7065	-
QUI	0	0.00	0.029210	2.1123	25.7392	513.3696	0.0000	7.7985	0.6937	-

--- Lines E/F: thermal shield supply / return ---
E: D = 110.3 mm (DN100, EN OD 114.3 x 2.0) m QUI = 1271.2 g/s dP = 376.485 mbar v_max = 7.10 m/s
F: D = 110.3 mm (DN100, EN OD 114.3 x 2.0) m QUI = 1271.2 g/s dP = 530.319 mbar v_max = 10.27 m/s

Line E key nodes										
Node	idx	x [m]	P [bar]	T [K]	h [J/g]	m [g/s]	dm_node [g/s]	v [m/s]	rho [kg/m^3]	flash [%]
RM	78	1039.37	18.000000	45.0000	250.4539	0.0000	0.0000	0.0000	18.3971	-
RM-1	77	1027.04	18.000000	45.0000	250.4539	26.8996	26.8996	0.1530	18.3971	-
RM-2	76	1014.71	18.000005	44.9936	250.4195	53.7992	26.8996	0.3060	18.3997	-
400 first	45	632.40	18.044921	44.9731	250.3143	887.6870	26.8996	5.0348	18.4518	-
800 last	44	619.16	18.049505	44.9728	250.3132	896.4032	8.7162	5.0830	18.4564	-
MID	39	552.98	18.073777	44.9714	250.3078	939.9839	8.7162	5.3231	18.4806	-
QUI+2	2	63.24	18.332409	44.9605	250.2740	1262.4816	8.7162	7.0514	18.7373	-
800 first	1	50.00	18.341538	44.9602	250.2732	1271.1977	8.7162	7.0967	18.7463	-
QUI	0	0.00	18.376485	44.9590	250.2703	1271.1977	0.0000	7.0836	18.7809	-

Line F key nodes										
Node	idx	x [m]	P [bar]	T [K]	h [J/g]	m [g/s]	dm_node [g/s]	v [m/s]	rho [kg/m^3]	flash [%]
RM	78	1039.37	16.000000	55.0000	303.3729	0.0000	0.0000	0.0000	13.4986	-
RM-1	77	1027.04	16.000000	55.0000	303.3729	26.8996	26.8996	0.2086	13.4986	-
RM-2	76	1014.71	15.999993	55.1385	304.1054	53.7992	26.8996	0.4181	13.4654	-
400 first	45	632.40	15.938276	55.2701	304.7908	887.6870	26.8996	6.9411	13.3841	-
800 last	44	619.16	15.931958	55.2766	304.8241	896.4032	8.7162	7.0128	13.3774	-
MID	39	552.98	15.898418	55.3073	304.9811	939.9839	8.7162	7.3727	13.3430	-
QUI+2	2	63.24	15.533233	55.4733	305.7987	1262.4816	8.7162	10.1563	13.0091	-
800 first	1	50.00	15.520084	55.4767	305.8148	1271.1977	8.7162	10.2354	12.9977	-
QUI	0	0.00	15.469681	55.5021	305.9405	1271.1977	0.0000	10.2722	12.9511	-